

QUENTIN BACQUELÉ

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 GitHub |  Google Scholar

PhD candidate in acoustic biogeography. I believe bioacoustics should be scaled to the biogeographic level, uncovering patterns that remain invisible at local scales. The rise of machine learning in bioacoustics, combined with vast amounts of available data and techniques from spatial, phylogenetic, and functional ecology, is opening the door to a new macroecological framework for studying acoustic communication.

EDUCATION

PhD Candidate in Acoustic Communication, Biogeography & AI <i>EPHE / ENES / CEFE</i>	Saint-Étienne, France 2024 – Present
M.Sc. International Master in Bioacoustics (MoBi) <i>Jean Monnet University</i>	Saint-Étienne, France 2023 – 2024
M.Sc. Ecophysiology, Ecology, and Ethology (EEE) <i>Strasbourg University – Double Degree</i>	Strasbourg, France 2021 – 2024
M.Sc. Interdisciplinary Approach in Data Science (AISD) <i>Strasbourg University – Double Degree</i>	Strasbourg, France 2021 – 2024
University Degree in Bioacoustics <i>Jean Monnet University</i>	Saint-Étienne, France 2023
B.Sc. in Life Sciences, specializing in Ecology <i>Savoie Mont-Blanc University</i>	Chambéry, France 2020 – 2021

PUBLICATIONS

HAC-Net: Learning Natural Units from Acoustic Change Quentin Bacquéle, Jean-Yves Barnagaud, Frederic Theunissen, Nicolas Mathevon	2025 NeurIPS 2025
AI Decodes Chimpanzee Vocabulary Items Antoine Valet, Quentin Bacquéle, Cédric Girard-Buttoz, Roman M Wittig, Catherine Crockford	2025 NeurIPS 2025
A novel contour-based approach for quantifying cryptic vocal dialects Quentin Bacquéle, Barbara Freitas, Borja Milá, Christophe Thébaud, et al.	In Prep
The global biogeography of passerine songs Quentin Bacquéle, Jean-Yves Barnagaud, Cyrille Violle, et al.	In Prep

RESEARCH EXPERIENCE

A novel contour-based approach for quantifying cryptic vocal dialects <i>University of California at Berkeley – Supervised by F. Theunissen & N. Mathevon</i>	Berkeley, USA 2024
Can chimpanzees flexibly combine calls to create new meanings? <i>ENES Lab, Jean Monnet University – Supervised by F. Levrero, C. Crockford & C. Girard-Buttoz</i>	Saint-Étienne, France 2023
An AI-based approach for planarian behavior analysis <i>Institute of Cellular and Integrative Neuroscience (CNRS) – Supervised by H. Cadiou</i>	Strasbourg, France 2023
Passive Acoustic Monitoring of wild bottlenose dolphins <i>Bottlenose Research Institute (BDRI) – Supervised by S. Methion & B. Diaz Lopez</i>	Ría de Arousa, Spain 2022

PRESENTATIONS

HAC-Net: Learning Natural Units from Acoustic Change <i>NeurIPS 2025 – Poster Presentation</i>	San Diego, USA Dec 2025
An AI based approach for planarian behavior analysis <i>NeuroFrance 2023 – Poster Presentation</i>	Lyon, France 2023
Passive Acoustic Monitoring to discover the hidden nightlife of dolphins <i>34th European Cetacean Society Conference – Poster Presentation</i>	O Grove, Spain 2023

GRANTS & FELLOWSHIPS

PhD Stipend (PSL-EPHE EABIS Grant) – €80,400 <i>University Paris-Sciences-Lettres</i>	2024 – 2027
Mobility Grant – €500 <i>EPHE</i>	2024

TEACHING

Sound Resynthesis (3h) <i>Bioacoustics Winter School</i>	Jean Monnet University 2026
Sound Resynthesis (4h) <i>M.Sc. Bioacoustics</i>	Jean Monnet University 2025
Statistics (16h) <i>Master 2 Ecology</i>	Jean Monnet University 2024 – 2025
Ecology - Practical Work (28h) <i>B.Sc. Life Sciences (L2 & L3)</i>	Jean Monnet University 2024

MENTORSHIP & SUPERVISION

Python for Bioacoustics <i>Mentorship of 2 students</i>	UC Berkeley 2024
M.Sc. Project Co-Supervisor <i>M.Sc. in Bioacoustics (MoBi)</i> • Topic: Biogeography of rhythm in passerine birds • Topic: Phylogeny of non-linear phenomena in mammals	Jean Monnet University 2024

OTHER EXPERIENCE

FauNet (Founder) <i>Client: SILABE – Facial recognition of rhesus monkeys</i>	Strasbourg, France 2022 – 2024
Competitions <i>Automatic Detection of Odontocetes' Clicks (Toulon University)</i>	Online 2023